

Job-retention Schemes and Employment During Covid: Evidence From Albion

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Abstract

We examine the determinants and employment effects of a Covid-era job-retention loan scheme in Albion. Using matching and difference-in-differences, we find that the scheme had a positive and significant effect on employment.

Keywords: Loan, Employment, difference-in-differences, Matching.

1 Introduction

The COVID-19 shock severely disrupted the economy, threatening firm survival and employment ([International Labour Office, 2020](#)). To prevent deeper economic collapse, many countries introduced job-retention schemes. This was the case in Albion, a country comprising a northern and a southern state. In 2020, the northern state introduced a job-retention scheme providing a five-year loan equal to 20% of a firm's total payroll. While all applicants received the funds, many eligible firms chose not to apply. Conversely, concerned about the expense, the southern state did not implement a similar programme.

This setting raises questions regarding the efficiency of such policies and their specific impact on firm employment. While extensive literature finds these interventions efficient ([Calligaris et al., 2023](#); [Lam and Solovyeva, 2023](#)), other studies suggest their employment effects were limited in certain contexts ([Granja et al., 2022](#); [Chetty et al., 2024](#)). Consequently, it is relevant to analyse the scheme's employment effects within Albion's northern state. Furthermore, the variation in loan take-up suggests that participation is driven by specific firm characteristics. Identifying these predictors could help policymakers better target future crisis support. Accordingly, this paper examines the determinants of loan receipt before estimating the policy's causal effect on firm employment.

2 Data

Table 1: Descriptive statistics by region and treatment status

Variable	By region			By treatment status				
	North	South	Diff	Treated	Untreated N	Diff	All untreated	Diff
<i>Panel a: means</i>								
<i>N</i>	1500	1500		1050	450		1950	
Employment	10.42 (3.45)	7.26 (2.99)	3.16***	11.63 (3.00)	7.60 (2.72)	4.03***	7.34 (2.93)	4.29***
Leverage	0.404 (0.220)	0.410 (0.220)	-0.007	0.420 (0.220)	0.360 (0.230)	0.06***	0.400 (0.220)	0.02***
Age	16.48 (8.59)	16.37 (8.38)	0.114	16.58 (8.61)	16.26 (8.54)	0.32	16.34 (8.42)	0.24
<i>Panel b: employment over time</i>								
<i>N</i>	500	500		350	150		650	
Employment 2018	9.30 (3.31)	7.46 (3.03)	1.84***	10.47 (2.89)	6.57 (2.49)	3.90***	7.25 (2.94)	3.22***
Employment 2019	10.54 (3.18)	7.67 (2.96)	2.87***	11.54 (2.91)	8.21 (2.52)	3.33***	7.80 (2.87)	3.74***
Employment 2020	11.42 (3.53)	6.64 (2.88)	4.78***	12.87 (2.69)	8.03 (2.85)	4.85***	6.96 (2.93)	5.91***

Standard deviations in parentheses.

The first Diff reports the difference in means between North and South.

the second reports the difference between treated and untreated in the North, the third reports the difference between treated and all untreated firms.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Observable variables are : loan, employment, leverage, age, industry, year, id and state. The data form a panel of 1,000 random firms followed over three years (2018 to 2020). Firms are equally split between two regions. When the policy was introduced in the North in 2020, 350 firms applied for and received a loan, while 150 did not apply. Throughout, “untreated” will refer to all untreated firms (in north and south).

We first use t-tests of mean differences, which are mainly descriptive (De Boer et al., 2015). The naive difference in 2020 employment between treated firms and untreated firms is 5.91, but this is unlikely to reflect the true causal effect because it ignores pre-existing differences. The table also allows us to compute a simple DiD estimate, as a first comparative benchmark. Comparing the change in employment from 2019 to

2020 for treated firms and untreated firms gives $(12.87 - 11.54) - (6.96 - 7.80) = 2.17$. Relative to untreated firms, treated firms experienced an additional increase of 2.17 employees between 2019 and 2020.

Panel a shows significant differences in pooled means of employment between the North and the South. Since the policy was introduced only in the North in 2020, and only 70% of northern firms received the loan, treated observations account for only a limited share of the full panel. This makes the loan effect less visible, especially because panel b reports differences by region and treatment status over all years. Leverage follows has significant difference both across across treatment status. As leverage, defined as liabilities over assets, proxies firm financial health (Bernanke et al., 1999), it may reflect a selection in firms who applied to the loan. By contrast, age is not statistically significant across regions or treatment status.

Panel b also points to structural differences between regions. Employment in the North increased in 2019 and 2020, whereas in the South it rose slightly in 2019 before falling sharply in 2020 (13.4%). The same pattern appears by treatment status: the treated group maintained a positive trend despite Covid in 2020, while the untreated group broke this trend, with employment declining by 10.8%. This may suggest that the policy helped support the North, especially treated firms, but it may also reflect structural regional differences or selection into treatment, calling for deeper analysis.

3 Methodology

3.1 Selection Bias and Endogeneity

Selection bias is evident because treated and control firms in the north exhibit pre-existing significant differences in leverage and employment. Firms self-select into the loan program, though the direction of this bias is ambiguous: both vulnerable firms anticipating severe pandemic impacts and financially stable firms confident in repayment are likely to apply. The economic logic and the evidence discussed below lead us to favor the vulnerability hypothesis.

Descriptive evidence suggests that in the north firms with more employees were more exposed to the shock and thus more likely to benefit from the policy. Higher leverage among treated firms may also have raised treatment probability. Firms may therefore differ in both observed and unobserved characteristics.

This self-selection introduces endogeneity. Firms differ in unobserved fixed effects (policies, history) and idiosyncratic characteristics (market confidence, profits, crash risk), and some may have used the policy simply to fund pre-existing hiring plans. Consequently, the policy becomes correlated with the error term, violating the Conditional Independence Assumption (CIA) and rendering the OLS estimator biased and inconsistent (Stock and Watson, 2020). Therefore, a good estimator should address this issue.

3.2 Prediction of loan receipt

We restrict our sample to northern firms in 2020, where the policy was implemented. Since all applicants received the loan, our dependent variable, a dummy equal to one if the loan was received in 2020, effectively models the firm’s participation decision.

Explanatory variables include early-2020 leverage, age, and industry, alongside 2019 employment. We use 2019 employment rather than 2020 to prevent post-treatment bias, as year-end 2020 figures could already reflect the loan’s impact. This ensures all explanatory variables are strictly pre-treatment.

We exclude temporal changes in leverage or employment to avoid overfitting and multicollinearity. Because these variables are highly persistent, their level at time t sufficiently captures both initial conditions and historical trajectories. Industry is controlled for using two dummy variables, with agriculture serving as the reference category.

Given the binary dependent variable, we estimate probit and logit models, reporting average marginal effects (AME). Following an underlying latent variable framework (Wooldridge, 2002), we interpret the probit’s latent variable as the firm’s unobserved net benefit or propensity to take the loan, with participation occurring if this benefit exceeds a specific threshold. We also estimate a Linear Probability Model (LPM) as a benchmark aware that it is internally inconsistent. We account for the inherent heteroskedasticity of its error term by applying robust standard errors.

$$\text{Let : } X_i = (1, \text{leverage}_{i,2020}, \text{employment}_{i,2019}, \text{age}_{i,2020}, \text{manufacturing}_{i,2020}, \text{services}_{i,2020})',$$

The coefficient vector is denoted by : $\beta = (\alpha, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5)'$.

The LPM is written as :

$$\text{loan}_{i,2020} = \alpha + \beta_1 \text{leverage}_{i,2020} + \beta_2 \text{employment}_{i,2019} + \beta_3 \text{age}_{i,2020} + \beta_4 \text{manufacturing}_{i,2020} + \beta_5 \text{services}_{i,2020} + u_i.$$

Under the LPM, $\mathbb{E}(y_i | X_i) = X_i' \beta$ and therefore $\Pr(\text{loan}_{i,2020} = 1 | X_i) = X_i' \beta$.

The Logit and Probit specifications are given by $\Pr(\text{loan}_{i,2020} = 1 | X_i) = F(X_i' \beta)$. We do not elaborate further on theoretical properties of logit and probit.

3.3 Causal effect of loan on employment

We evaluate three methods. Guided by the predictive results of loan receipt, we retain two approaches for further analysis and ultimately select the most robust causal estimator.

Instrumental variables (IV) : it can address endogeneity (Cameron and Trivedi, 2005). An indicator for being in the North in 2020 can be an instrument for loan receipt. This strategy relies on four assumptions. Relevance is plausible, since firms in the North in 2020 were more likely to receive the loan. Monotonicity is also plausible if the policy only encouraged participation. However, the exclusion restriction is doubtful,

because the North may have faced region-specific shocks in 2020 that affected employment directly. Exogeneity is also weak, as regional differences observed in description of covariates suggest deeper structural differences in unobserved determinants of employment. We therefore do not retain IV.

Panel data: Observing the same firms from 2018 to 2020 allows us to use panel data methods, such as FE and FD, to control for unobserved, time-invariant firm characteristics. However, these methods do not fully resolve endogeneity, as causal interpretation still assumes treatment is uncorrelated with time-varying unobservables (Cameron and Trivedi, 2005). In our setting, selection bias likely violates this assumption: firms anticipating a steeper decline in employment may have been more likely to seek the 2020 policy support. Consequently, loan receipt would remain correlated with time-varying idiosyncratic shocks, rendering it endogenous despite accounting for time-invariant heterogeneity. Therefore, we mainly utilize fixed effects as a robustness check and for cross-estimator comparison.

$$\begin{aligned} \text{FE:} \quad \text{employ}_{it} - \bar{\text{employ}}_i &= \beta_1(\text{loan}_{it} - \bar{\text{loan}}_i) + \beta_2(\text{leverage}_{it} - \bar{\text{leverage}}_i) + \beta_3(\text{age}_{it} - \bar{\text{age}}_i) \\ &\quad + (\lambda_t - \bar{\lambda}_i) + (\varepsilon_{it} - \bar{\varepsilon}_i) \end{aligned}$$

The Difference-in-Difference : This approach is highly suitable given our panel structure, which tracks distinct treated and control groups across two pre-policy periods and one post-policy period. The core identifying assumption is that these groups would have followed parallel trends in the absence of the intervention.

Subsample Selection : Comparing treated and untreated firms within the North controls for region-specific shocks. However, this strategy relies on the parallel trends assumption, which is difficult to justify given the selection into treatment. Furthermore, even if pre-2020 trends were parallel, underlying differences in firm characteristics and resilience mean the two groups likely responded differently to the simultaneous COVID-19 shock. Consequently, historical parallel trends alone are insufficient for identification when the policy coincides with such a major shock.

Comparing the North overall to the South provides a stable control group but dilutes the causal effect, as the northern sample includes both treated and untreated firms. Comparing treated northern firms to all untreated firms partially addresses this dilution. However, the ideal counterfactual (southern firms that would have applied) is unobservable.

DiD assumes common time effects (Cameron and Trivedi, 2005) and likely underestimates the true causal impact due to selection bias : more vulnerable firms selected into the program, they would have faced a steeper employment decline in the absence of the policy. Consequently, if post-shock paths are similar across groups, a naive DiD estimate may understate the true effect. A more reliable estimator could reveal a larger causal effect relative to the DiD estimate.

The DiD equation is :

$$\Delta \text{employment}_{it} = \delta + \beta_{ATET} \Delta \text{loan}_{it} + \Delta \varepsilon_{it}$$

These limitations motivate combining DiD with matching. As our prediction model and (Cameron and Trivedi, 2005) suggest, matching is effective here since descriptive analysis shows selection into treatment by observable characteristics.

Restricting controls to the North yields only 150 controls for 350 treated firms, which would necessitate repeated matches and risk generating unstable estimates. The broader pool allows the matching algorithm to identify observable similar firms regardless of region, partially offsetting the weakness of regional heterogeneity. Thus, we consider all untreated firms.

The overlap assumption appears plausible, since each firm has a positive probability of remaining untreated and the groups are broadly comparable in observed characteristics. However, the CIA may be strained by our limited set of covariates. We therefore assume that relevant unobserved characteristics correlate with observed ones. This limitation is mitigated by layering matching with the DiD.

Following Dehejia and Wahba (1999), we match exclusively on pre-treatment information: 2019 employment and early-2020 values for the remaining variables. This ensures covariates are uncontaminated by the policy. For example, treated firms' leverage barely shifted between 2019 and early 2020 (from 0.424 to 0.427), confirming it does not reflect the loan influx. Ultimately, we estimate the ATET using a nearest-neighbour matching DiD estimator:

$$\hat{\beta}_{ATET}^{match} = \frac{1}{N_1} \sum_{i:treated_north_i=1} \left[change_employment_i - \sum_{j:treated_north_j=0} w_{ij} change_employment_j \right]$$

where w_{ij} is the weight assigned to untreated j in the counterfactual for treated i .

In summary, while standard panel methods help us explore the role of FE, we ultimately rely on a matched DiD to isolate the causal effect.

4 Results

4.1 Prediction of loan receipt

We report AME rather than marginal effects at the mean, as the latter can be misleading in the presence of categorical variables. We present it in appendix (7) for robustness.

Table 2: determinants of loan receipt

Covariates	LPM ^a	Logit ^b	Probit ^b
Leverage	0.246*** (0.073)	0.234*** (0.072)	0.230*** (0.072)
Employment ₂₀₁₉	0.077*** (0.006)	0.074*** (0.005)	0.074*** (0.005)
Age	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)
Industry 2	-0.105 (0.074)	-0.105* (0.058)	-0.099* (0.058)
Industry 3	-0.019 (0.072)	-0.016 (0.056)	-0.015 (0.057)
Constant	4.8 × 10 ⁻⁴ (0.092)		
Observations	500	500	500
R-squared	0.283	-	-
Pseudo R-squared	-	0.270	0.271
F statistic / lr chi ²	46.60***	164.77***	165.60***
Log likelihood	-	-223.05	-222.63

Dependent variable = loan in 2020; SE in parentheses.

^a we use robust se for LPM

^b coefficients for logit and probit are AME.

*** p<0.01, ** p<0.05, * p<0.10.

Logit and probit coefficients (Appendix Table 6) reveal that leverage, 2019 employment, and firm age are statistically significant predictors of loan receipt. Findings are robust, remaining stable across the LPM, logit, and probit specifications. We rely on logit because it is the standard model for propensity score matching for consistency with the next sections. Manufacturing (Industry 2) exhibits only weak significance (10% level). Covariates are jointly significant across all specifications, confirmed by F-statistics in the LPM and Likelihood-Ratio tests in the logit and probit models. A one-unit increase in leverage raises the probability of loan receipt by 23.4 percentage points (*ceteris paribus*), underscoring non-random selection where heavily indebted firms are more likely to seek support. Similarly, each additional employee in 2019 increases treatment probability by 7.4 percentage points, likely reflecting the payroll pressures associated with a larger workforce. Conversely, each additional year of firm age reduces the likelihood of loan receipt by 0.9 percentage points, suggesting that older firms possess greater financial resilience.

Firm 453 exemplifies these dynamics, exhibiting the sample's highest predicted probability of treatment.

Although its relatively advanced age (25 years) exerts a downward pressure on probability, this is overwhelmingly offset by its high leverage (0.713, 90th percentile) and employment (19 workers, upper tail), driving its high propensity for loan uptake.

4.2 Causal effect of loan on employment : Difference In Difference and Matching

4.2.1 Matching

We match on cross-sectional of variables in 2020 apart from the employment for which we considered the 2019 values.

Table 3: matching estimates of the loan effect on employment

	North : treated vs untreated		North treated vs All untreated	
	Ma	PS	Ma	PS
Atet on employment ₂₀₂₀	3.926*** (0.412)	3.377*** (0.511)	4.700*** (0.287)	4.361*** (0.378)
Z-statistic	9.53	6.61	16.40	11.54
Total observations	500	500	1000	1000
Treated observations	350	350	350	350
Control observations	150	150	650	650
Matched control observations	350	350	350	350

ma = nearest-neighbour matching based on mahalanobis distance.

ps = propensity-score matching based on a logit model.

reported effects are atet estimates. ai-robust standard errors are reported in parentheses.

*** significance at the 1% .

“all untreated” includes untreated firms in both the north and the south.

Mahalanobis matching is appropriate here because the covariates are measured on different scales and are not too many. Controlling for firm characteristics yields a larger employment effect than the DiD (2.17). Furthermore, the choice of control group is consequential: matching with North-only controls estimates an effect of 3.926, which increases to 4.7 in all sample. Consistent with our previous discussion that severe selection bias within the North substantially underestimates the true causal effect.

Figure 1: Quality of match employment

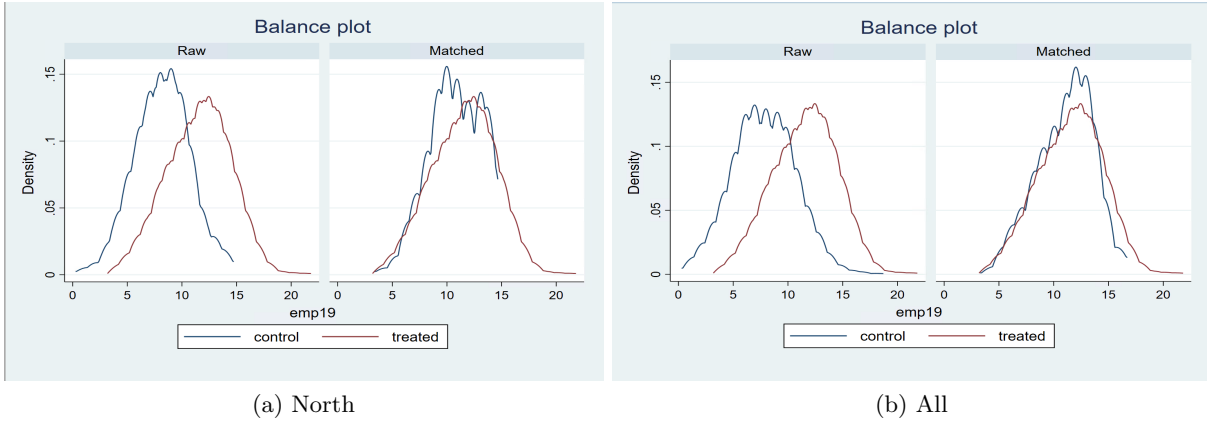
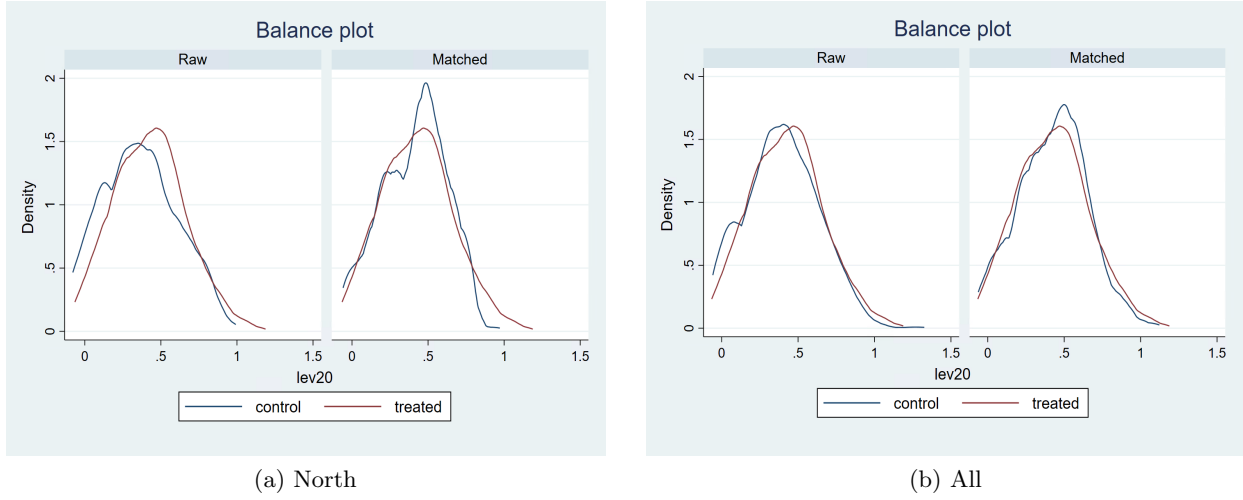


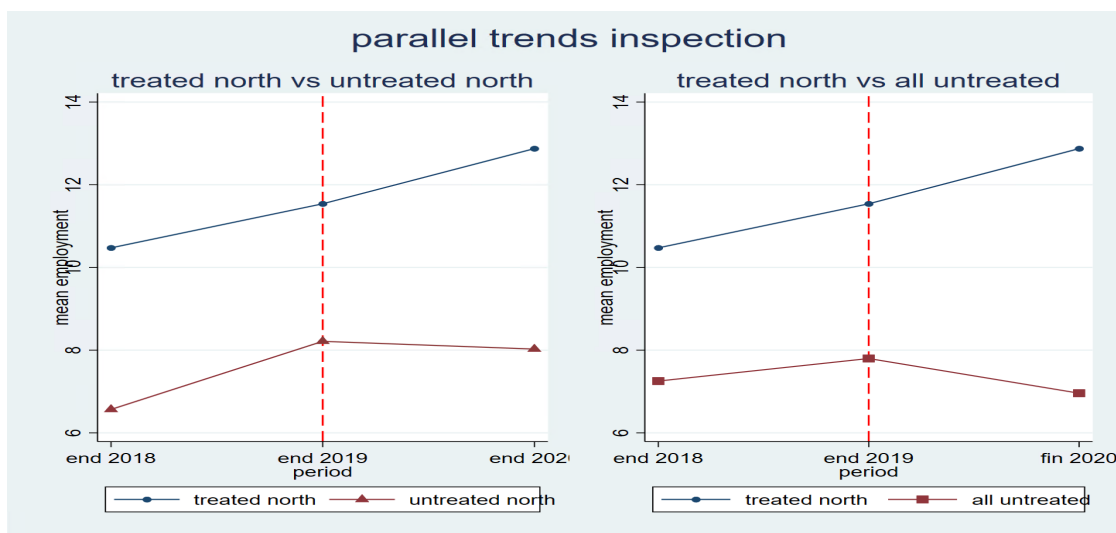
Figure 2: Quality of match leverage



These figures demonstrate weaker match quality for employment and leverage on the left compared to the right. This confirms that a larger pool of control observations increases overlap and significantly improves matching. Consequently, we utilize the full untreated sample. Match quality for firm age is reported in [Appendix 5](#).

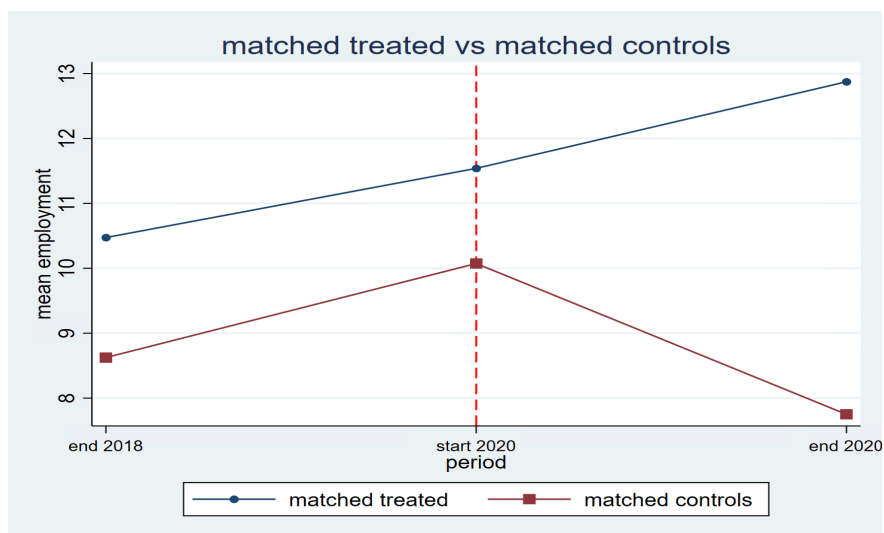
4.2.2 Combination of Difference-in-Differences and matching

Figure 3: Parallel trends before the match



The parallel trends between treated and untreated firms in the North appear reasonably close, although not perfect. Before the shock, both had a positive trend, after the shock the treated maintained the trend but the untreated experienced a decrease in employment. As discussed above, we cannot rely solely on the visual similarity of these trends, given the presence of the Covid shock and the likely selection bias. By contrast, when the comparison group is expanded to include all untreated firms in the sample, the parallel trends assumption appears much less plausible.

Figure 4: Parallel trends after the match



While perfectly parallel trends are empirically rare in finite samples, matching on pre-treatment characteristics strengthens the parallel trends assumption. The matched counterfactual exhibits a much steeper

employment decline than the previous one. This indicates that observationally equivalent untreated firms suffered severe job losses, implying the true causal effect is substantially larger than the naive DiD estimate.

Table 4: estimated effect of the loan on employment

	DiD	DiD + leverage control	DiD + matching
Loan / treated effect	2.174*** (0.195)	2.188*** (0.195)	4.240*** (0.308)
Observations	1000	1000	1000
R^2	0.111	0.113	

standard errors in parentheses. the third column reports the atet from nearest-neighbour mahalanobis matching with one neighbour.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Introducing leverage into the standard DiD increases the estimated effect, supporting that the baseline model underestimates the true impact ignoring firm characteristics. Our preferred estimator (DiD combined with matching) yields an ATET of 4.24 jobs, approximately double the baseline DiD. This approach is optimal because matching addresses observable selection bias while DiD removes common time trends.

4.2.3 Robustness : Panel Data

In our panel, we control for year effects.

Table 5: panel estimators of employment

variables	Pols	FE	FD
Loan	5.400*** (0.200)	2.447*** (0.168)	2.186*** (0.193)
Leverage	-0.184 (0.248)	-1.147* (0.589)	-1.213* (0.673)
Age	0.100*** (0.006)	-0.101* (0.055)	
Industry = 2	1.424*** (0.156)		
Industry = 3	2.390*** (0.160)		
Year = 2019	0.629*** (0.133)	0.837*** (0.085)	
Year = 2020	-1.441*** (0.151)		-1.584*** (0.147)
Constant	5.345*** (0.208)	10.410*** (0.916)	0.736*** (0.092)
Observations	3000	3000	2000
R^2	0.320	0.127	0.078

SE in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

R^2 for the FE is the within R^2 .

FE and FD omit time-invariant variables.

In both the FE and FD models, the estimated effect of the loan remains positive and statistically significant. In the FE specification, receiving the loan is associated with an increase of 2.45 employees, slightly larger than in the FD (2.18). This may be because FE uses all three periods allows control for more firm observations, which reduce selection bias than FD, which relies only on changes between the last two periods. This is consistent with our previous findings.

Both estimates are close to the DiD results. In particular, the FD estimate, which controls for time effects and leverage, is identical to the DiD estimate with leverage controls. This shows that since both models do not fully address selection bias, they yields in the same underestimated coefficient.

This confirms that DiD and matching were relevant for obtaining a more credible estimate.

5 Conclusion

Our findings suggest that weaker firms were more likely to take up the loan in order to prevent a deeper decline during Covid. Firms with higher debt relative to assets and those with higher employment were more likely to participate, while older firms were less likely to do so. Overall, the results suggest that the job-retention scheme had a positive causal effect on employment. Selection bias appears to have led the panel and DiD estimators to underestimate this effect. Once we control for firm characteristics through matching and for time effects through DiD, we obtain a positive and statistically significant estimated causal effect of the loan on employment of 4.24 jobs per firm on average, which is more likely to be close to the true effect.

The limited number of covariates and the lack of precision regarding the extent of homogeneity between the two regions remain important limitations of our analysis.

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6 Appendix

Table 6: coefficient estimates for loan receipt models

covariates	lpm ^a	logit ^b	probit ^b
leverage	0.246*** (0.073)	1.618*** (0.512)	0.922*** (0.295)
employment ₂₀₁₉	0.077*** (0.006)	0.513*** (0.054)	0.297*** (0.029)
age	-0.009*** (0.002)	-0.061*** (0.016)	-0.035*** (0.009)
industry 2	-0.105 (0.074)	-0.716* (0.423)	-0.394 (0.245)
industry 3	-0.019 (0.072)	-0.122 (0.421)	-0.065 (0.245)
constant	4.8×10^{-4} (0.092)	-3.425*** (0.609)	-1.983*** (0.344)
observations	500	500	500
r-squared	0.283	-	-
pseudo r-squared	-	0.270	0.271
f statistic / lr chi ²	46.60***	164.77***	165.60***
log likelihood	-	-223.05	-222.63

dependent variable = loan receipt in 2020.

standard errors in parentheses.

^a lpm estimated by ols with robust standard errors.

^b logit and probit columns report raw coefficient estimates.

base category for industry is industry 1.

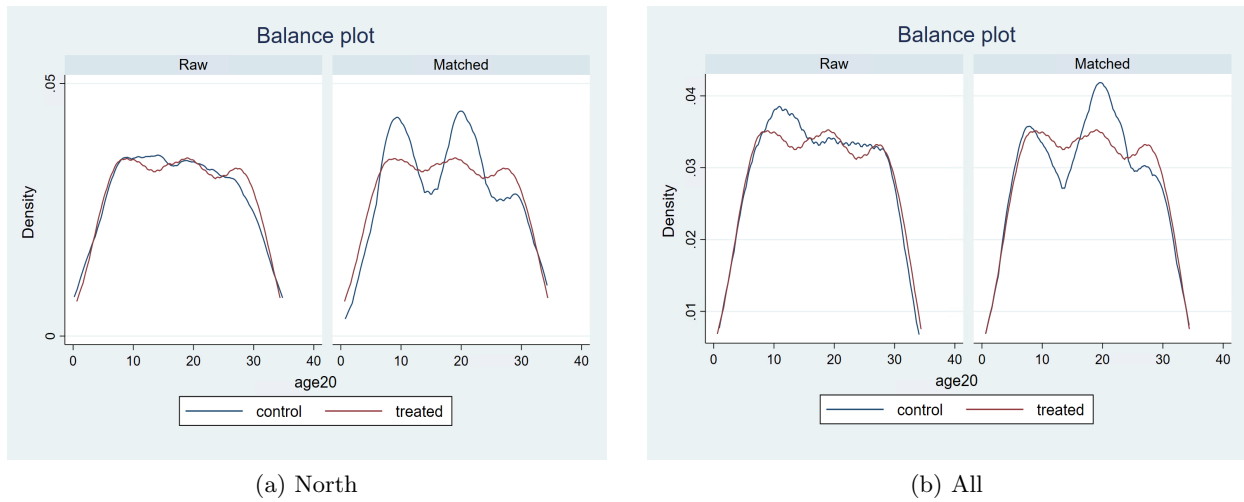
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7: robustness check: average and marginal effects at means

Covariates	logit		probit	
	Ame	Mem	Ame	Mem
Leverage	0.234***	0.283***	0.230***	0.282***
Employment ₂₀₁₉	0.074***	0.090***	0.074***	0.091***
Age	-0.009***	-0.011***	-0.009***	-0.011***
Industry 2	-0.105*	-0.126*	-0.099*	-0.121*
Industry 3	-0.016	-0.018	-0.015	-0.018
Observations	500	500	500	500

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 5: Quality of match age



RAW OUTPUTS AND STATA COMMANDS

DESCRIPTIVE TABLE 1

```
. do "C:\Users\qclm20\AppData\Local\Temp\STD1c20_000000.tmp"

. * Test Stat
. *North-South
. ttest employment, by(state)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
0	1,500	7.256667	.0772273	2.991001	7.105182	7.408152
1	1,500	10.42133	.0890983	3.450761	10.24656	10.5961
Combined	3,000	8.839	.0656457	3.59556	8.710285	8.967715

```
diff |          -3.164667   .1179091          -3.395858   -2.933476

diff = mean(0) - mean(1)                                t = -26.8399
H0: diff = 0                                             Degrees of freedom =   2998

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000          Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000

. ttest age, by(state)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
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```

0 | 1,500      16.37   .2163802   8.380369   15.94556   16.79444
1 | 1,500      16.484   .2218071   8.590553   16.04891   16.91909
-----+-----
Combined | 3,000      16.427   .1549122   8.484889   16.12326   16.73074
-----+-----
diff |           -.114    .309869           -.7215774   .4935774
-----+-----
diff = mean(0) - mean(1)                                t =  -0.3679
H0: diff = 0                                           Degrees of freedom = 2998

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.3565          Pr(|T| > |t|) = 0.7130          Pr(T > t) = 0.6435

```

```
. ttest leverage, by(state)
```

Two-sample t test with equal variances

```

-----+-----
Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |    1,500   .4116572   .0055911   .2165404   .4006901   .4226243
1 |    1,500   .4042757   .0057019   .220833   .3930912   .4154602
-----+-----
Combined | 3,000   .4079664   .0039928   .218692   .4001376   .4157952
-----+-----
diff |           .0073815   .0079857           -.0082765   .0230395
-----+-----
diff = mean(0) - mean(1)                                t =   0.9243
H0: diff = 0                                           Degrees of freedom = 2998

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.8223          Pr(|T| > |t|) = 0.3554          Pr(T > t) = 0.1777

```

```
. *North Treated-Untreated
```

```
. ttest employment if state==1, by(treated2020)
```

Two-sample t test with equal variances

```

-----+-----
Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |      450   7.602222   .1283747   2.723239   7.349932   7.854512
1 |    1,050  11.62952   .0924342   2.995211  11.44815  11.8109
-----+-----
Combined | 1,500  10.42133   .0890983   3.450761  10.24656  10.5961
-----+-----
diff |      -4.027302   .164318           -4.349619  -3.704984
-----+-----
diff = mean(0) - mean(1)                                t = -24.5092
H0: diff = 0                                           Degrees of freedom = 1498

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000          Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000

```

```
. ttest age if state==1, by(treated2020)
```

Two-sample t test with equal variances

```
-----+-----
Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |      450      16.26   .4026427    8.541342    15.4687    17.0513
1 |     1,050      16.58   .2658286    8.61383    16.05838    17.10162
-----+-----
Combined |    1,500      16.484   .2218071    8.590553    16.04891    16.91909
-----+-----
diff |           -.32   .4841138           -1.269613    .6296129
-----+-----
diff = mean(0) - mean(1)                                t =  -0.6610
H0: diff = 0                                           Degrees of freedom =   1498

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.2544          Pr(|T| > |t|) = 0.5087          Pr(T > t) = 0.7456
```

. ttest leverage if state==1, by(treated2020)

Two-sample t test with equal variances

```
-----+-----
Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |      450   .362956   .0106482   .2258825   .3420295   .3838825
1 |     1,050   .4219841   .0066763   .2163375   .4088837   .4350846
-----+-----
Combined |    1,500   .4042757   .0057019   .220833   .3930912   .4154602
-----+-----
diff |           -.0590281   .0123529           -.0832589   -.0347973
-----+-----
diff = mean(0) - mean(1)                                t =  -4.7785
H0: diff = 0                                           Degrees of freedom =   1498

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000          Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000
```

. *North Treated vs North&South Untreated

. ttest employment, by(treated2020)

Two-sample t test with equal variances

```
-----+-----
Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |     1,950    7.33641   .0664492    2.934317    7.206091    7.466729
1 |     1,050   11.62952   .0924342    2.995211   11.44815   11.8109
-----+-----
Combined |    3,000     8.839   .0656457    3.59556    8.710285    8.967715
-----+-----
diff |           -4.293114   .1131407           -4.514955   -4.071272
-----+-----
diff = mean(0) - mean(1)                                t = -37.9449
H0: diff = 0                                           Degrees of freedom =   2998

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
```

Pr(T < t) = 0.0000 Pr(|T| > |t|) = 0.0000 Pr(T > t) = 1.0000

. ttest age, by(treated2020)

Two-sample t test with equal variances

```
-----+-----
Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |    1,950   16.34462   .1905784   8.415713   15.97086   16.71837
1 |    1,050    16.58    .2658286   8.61383   16.05838   17.10162
-----+-----
Combined |   3,000    16.427   .1549122   8.484889   16.12326   16.73074
-----+-----
diff |           -.2353846   .3248099           -.8722574   .4014881
-----+-----
diff = mean(0) - mean(1)                                t =  -0.7247
H0: diff = 0                                           Degrees of freedom =   2998
```

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = 0.2344 Pr(|T| > |t|) = 0.4687 Pr(T > t) = 0.7656

. ttest leverage, by(treated2020)

Two-sample t test with equal variances

```
-----+-----
Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |    1,950   .4004184   .0049737   .2196341   .390664   .4101728
1 |    1,050   .4219841   .0066763   .2163375   .4088837   .4350846
-----+-----
Combined |   3,000   .4079664   .0039928   .218692   .4001376   .4157952
-----+-----
diff |           -.0215657   .0083632           -.0379639   -.0051675
-----+-----
diff = mean(0) - mean(1)                                t =  -2.5786
H0: diff = 0                                           Degrees of freedom =   2998
```

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = 0.0050 Pr(|T| > |t|) = 0.0100 Pr(T > t) = 0.9950

. *North-South

. ttest employment if year==2018, by(state)

Two-sample t test with equal variances

```
-----+-----
Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |      500     7.458   .135607   3.032264   7.191569   7.724431
1 |      500     9.302   .1478151   3.305245   9.011583   9.592417
-----+-----
Combined |   1,000     8.38   .1044055   3.301591   8.175121   8.584879
-----+-----
diff |           -1.844   .2005955           -2.237637   -1.450363
-----+-----
```

```
diff = mean(0) - mean(1)                                t = -9.1926
H0: diff = 0                                             Degrees of freedom = 998
```

```
Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000          Pr(|T| > |t|) = 0.0000        Pr(T > t) = 1.0000
```

```
. ttest employment if year==2019, by(state)
```

Two-sample t test with equal variances

```
-----+-----
Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |      500      7.674   .1325843   2.964676   7.413508   7.934492
1 |      500     10.542   .1423833   3.183788   10.26226   10.82174
-----+-----
Combined |    1,000      9.108   .1072933   3.392911   8.897454   9.318546
-----+-----
diff |              -2.868   .1945549              -3.249784   -2.486216
-----+-----
```

```
diff = mean(0) - mean(1)                                t = -14.7413
H0: diff = 0                                             Degrees of freedom = 998
```

```
Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000          Pr(|T| > |t|) = 0.0000        Pr(T > t) = 1.0000
```

```
. ttest employment if year==2020, by(state)
```

Two-sample t test with equal variances

```
-----+-----
Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |      500      6.638   .1287753   2.879503   6.384991   6.891009
1 |      500     11.42   .157737   3.527107   11.11009   11.72991
-----+-----
Combined |    1,000      9.029   .1268     4.009768   8.780175   9.277825
-----+-----
diff |              -4.782   .2036272              -5.181587   -4.382413
-----+-----
```

```
diff = mean(0) - mean(1)                                t = -23.4841
H0: diff = 0                                             Degrees of freedom = 998
```

```
Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000          Pr(|T| > |t|) = 0.0000        Pr(T > t) = 1.0000
```

```
. *North Treated-Untreated
```

```
. ttest employment if year==2018 & state==1, by(treated2020)
```

Two-sample t test with equal variances

```
-----+-----
Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |     150     6.566667   .2034197   2.491372   6.164707   6.968627
1 |     350    10.47429   .154699   2.894154   10.17003   10.77855
-----+-----
```

```

Combined |      500      9.302   .1478151   3.305245   9.011583   9.592417
-----+-----
diff |      -3.907619   .2712776          -4.440609   -3.374629
-----+-----
diff = mean(0) - mean(1)                                t = -14.4045
H0: diff = 0                                           Degrees of freedom =    498

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000          Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000

```

```
. ttest employment if year==2019 & state==1, by(treated2020)
```

```
Two-sample t test with equal variances
```

```

Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |      150   8.213333   .2060791   2.523944   7.806118   8.620548
1 |      350   11.54     .1553241   2.905848   11.23451   11.84549
-----+-----
Combined |      500   10.542     .1423833   3.183788   10.26226   10.82174
-----+-----
diff |      -3.326667   .2729647          -3.862971   -2.790362
-----+-----
diff = mean(0) - mean(1)                                t = -12.1872
H0: diff = 0                                           Degrees of freedom =    498

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000          Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000

```

```
. ttest employment if year==2020 & state==1, by(treated2020)
```

```
Two-sample t test with equal variances
```

```

Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |      150   8.026667   .2328592   2.851931   7.566534   8.4868
1 |      350  12.87429     .1438739   2.691634   12.59132   13.15725
-----+-----
Combined |      500   11.42     .157737   3.527107   11.11009   11.72991
-----+-----
diff |      -4.847619   .267453          -5.373094   -4.322144
-----+-----
diff = mean(0) - mean(1)                                t = -18.1251
H0: diff = 0                                           Degrees of freedom =    498

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000          Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000

```

```
. *North Treated vs North&South Untreated
```

```
. ttest employment if year==2018, by(treated2020)
```

```
Two-sample t test with equal variances
```

```

Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----

```

```

-----+-----
0 |      650      7.252308      .1152641      2.93867      7.025972      7.478643
1 |      350      10.47429      .154699      2.894154      10.17003      10.77855
-----+-----
Combined |      1,000      8.38      .1044055      3.301591      8.175121      8.584879
-----+-----
diff |           -3.221978      .1938049           -3.60229      -2.841666
-----+-----
diff = mean(0) - mean(1)                                t = -16.6249
H0: diff = 0                                           Degrees of freedom =      998

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000          Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000

```

```
. ttest employment if year==2019, by(treated2020)
```

```
Two-sample t test with equal variances
```

```

-----+-----
Group |      Obs      Mean      Std. err.      Std. dev.      [95% conf. interval]
-----+-----
0 |      650      7.798462      .112811      2.876127      7.576943      8.01998
1 |      350      11.54      .1553241      2.905848      11.23451      11.84549
-----+-----
Combined |      1,000      9.108      .1072933      3.392911      8.897454      9.318546
-----+-----
diff |           -3.741538      .1913768           -4.117085      -3.365991
-----+-----
diff = mean(0) - mean(1)                                t = -19.5506
H0: diff = 0                                           Degrees of freedom =      998

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000          Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000

```

```
. ttest employment if year==2020, by(treated2020)
```

```
Two-sample t test with equal variances
```

```

-----+-----
Group |      Obs      Mean      Std. err.      Std. dev.      [95% conf. interval]
-----+-----
0 |      650      6.958462      .1149269      2.930073      6.732788      7.184135
1 |      350      12.87429      .1438739      2.691634      12.59132      13.15725
-----+-----
Combined |      1,000      9.029      .1268      4.009768      8.780175      9.277825
-----+-----
diff |           -5.915824      .1888843           -6.28648      -5.545168
-----+-----
diff = mean(0) - mean(1)                                t = -31.3198
H0: diff = 0                                           Degrees of freedom =      998

Ha: diff < 0                Ha: diff != 0                Ha: diff > 0
Pr(T < t) = 0.0000          Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000

```

```
.
end of do-file
.
```

LPM, LOGIT AND PROBIT

```
. sort id year

. by id: gen employment_2019 = employment[_n-1] if year == 2020 & year[_n-1] ==
> 2019
(2,000 missing values generated)

. keep if state == 1 & year == 2020
(2,500 observations deleted)

.
. *MPL
. regress loan leverage employment_2019 age i.industry, robust
```

```
Linear regression                               Number of obs   =       500
F(5, 494)           =       46.60
Prob > F            =       0.0000
R-squared           =       0.2829
Root MSE           =       .39042
```

```
-----+-----
|               Robust
loan | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-----+-----
leverage |   .2457943   .0728631     3.37  0.001   .1026346   .388954
employm~2019 | .0769403   .0055094    13.97  0.000   .0661155   .087765
age |   -.0092317   .0022069    -4.18  0.000   -.0135677   -.0048957
|
industry |
2 |   -.1047753   .0740553    -1.41  0.158   -.2502774   .0407268
3 |   -.0186006   .0723761    -0.26  0.797   -.1608035   .1236022
|
_cons |    .00048   .0923947     0.01  0.996   -.1810551   .182015
-----+-----
```

```
.
.
. *Logit
. logit loan leverage employment_2019 age i.industry, nolog

Logistic regression                               Number of obs =       500
LR chi2(5)      = 164.77
Prob > chi2     = 0.0000
Log likelihood = -223.04791                       Pseudo R2      = 0.2697
```

```
-----+-----
loan | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-----+-----
leverage |   1.617839   .5119727     3.16  0.002   .6143914   2.621288
employm~2019 | .5132631   .0537713     9.55  0.000   .4078733   .6186528
age |   -.0614062   .0155395    -3.95  0.000   -.091863   -.0309495
```

```

|
industry |
2 | -.7163304 .4225629 -1.70 0.090 -1.544539 .1118777
3 | -.1215784 .42092 -0.29 0.773 -.9465665 .7034097
|
_cons | -3.425377 .6089048 -5.63 0.000 -4.618809 -2.231946
-----

. margins, dydx(*)

Average marginal effects                                Number of obs = 500
Model VCE: OIM

Expression: Pr(loan), predict()
dy/dx wrt: leverage employment_2019 age 2.industry 3.industry

-----
|               Delta-method
|      dy/dx   std. err.      z    P>|z|    [95% conf. interval]
-----+-----
leverage |   .2343831   .0716019     3.27  0.001   .0940459   .3747203
employm~2019 | .0743586   .0048397    15.36  0.000   .064873   .0838442
age |   -.0088962   .0021296    -4.18  0.000   -.0130701   -.0047222
|
industry |
2 |   -.1050283   .0576666    -1.82  0.069   -.2180527   .0079961
3 |   -.0164977   .0563162    -0.29  0.770   -.1268754   .0938801
-----

Note: dy/dx for factor levels is the discrete change from the base level.

.
.
. *Probit
. probit loan leverage employment_2019 age i.industry, nolog

Probit regression                                Number of obs =    500
LR chi2(5)    = 165.60
Prob > chi2    = 0.0000
Log likelihood = -222.63452                        Pseudo R2    = 0.2711

-----
loan | Coefficient   Std. err.      z    P>|z|    [95% conf. interval]
-----+-----
leverage |   .9215231   .2949076     3.12  0.002   .3435149   1.499531
employm~2019 | .2974663   .028865    10.31  0.000   .2408919   .3540407
age |   -.0351184   .008778     -4.00  0.000   -.0523231   -.0179138
|
industry |
2 |   -.3935292   .2451938    -1.60  0.108   -.8741003   .0870419
3 |   -.0648288   .2449372    -0.26  0.791   -.5448969   .4152393
|
_cons |  -1.983143   .3439886    -5.77  0.000   -2.657348  -1.308938
-----

```



```
. margins, dydx(*)
```

Average marginal effects

Number of obs = 500

Model VCE: OIM

Expression: Pr(loan), predict()

dy/dx wrt: leverage employment_2019 age 2.industry 3.industry

Delta-method						
	dy/dx	std. err.	z	P> z	[95% conf. interval]	
leverage	.2297267	.0716192	3.21	0.001	.0893556	.3700977
employ~2019	.0741554	.0046992	15.78	0.000	.0649452	.0833657
age	-.0087547	.0020947	-4.18	0.000	-.0128602	-.0046492
industry						
2	-.0992833	.0581482	-1.71	0.088	-.2132517	.0146851
3	-.0152485	.0569085	-0.27	0.789	-.1267872	.0962902

Note: dy/dx for factor levels is the discrete change from the base level.

```
.
. *Predict
. quietly regress loan leverage employment_2019 age i.industry

. predict xb_lpm, xb

. predict pr_lpm
(option xb assumed; fitted values)

.
. quietly logit loan leverage employment_2019 age i.industry

. predict xb_logit, xb

. predict pr_logit, pr

.
. quietly probit loan leverage employment_2019 age i.industry

. predict xb_probit, xb

. predict pr_probit, pr

.
. summarize loan pr_lpm xb_logit pr_logit xb_probit pr_probit, sep(2)
```

Variable	Obs	Mean	Std. dev.	Min	Max
loan	500	.7	.4587165	0	1
pr_lpm	500	.7	.2439631	.0434265	1.388186
xb_logit	500	1.229846	1.629307	-3.156823	5.823289

```

pr_logit |          500          .7   .2554532   .0408233   .9970509
-----+-----
xb_probit |          500   .7278454   .9413554  -1.798488   3.382906
pr_probit |          500   .7029035   .252678   .0360499   .9996414

.
. *Margins at mean
. quietly logit loan leverage employment_2019 age i.industry

. margins, predict(pr) dydx(*) atmean

Conditional marginal effects                Number of obs = 500
Model VCE: OIM

Expression: Pr(loan), predict(pr)
dy/dx wrt:  leverage employment_2019 age 2.industry 3.industry
At: leverage      = .4080508 (mean)
employment_2019 = 10.542 (mean)
age              = 17.484 (mean)
1.industry       = .084 (mean)
2.industry       = .388 (mean)
3.industry       = .528 (mean)

-----+-----
|              Delta-method
|      dy/dx   std. err.      z    P>|z|    [95% conf. interval]
-----+-----
leverage |   .2831836   .0894118     3.17   0.002    .1079396   .4584276
employm~2019 | .0898406   .0087805    10.23   0.000    .0726312   .10705
age |  -.0107484   .0026974    -3.98   0.000   -.0160352   -.0054616
|
industry |
2 |  -.126315   .0655335    -1.93   0.054   -.2547583   .0021283
3 |  -.0180055   .0606737    -0.30   0.767   -.1369238   .1009128
-----+-----

Note: dy/dx for factor levels is the discrete change from the base level.

. quietly probit loan leverage employment_2019 age i.industry

. margins, predict(pr) dydx(*) atmean

Conditional marginal effects                Number of obs = 500
Model VCE: OIM

Expression: Pr(loan), predict(pr)
dy/dx wrt:  leverage employment_2019 age 2.industry 3.industry
At: leverage      = .4080508 (mean)
employment_2019 = 10.542 (mean)
age              = 17.484 (mean)
1.industry       = .084 (mean)
2.industry       = .388 (mean)
3.industry       = .528 (mean)

```



```

employ~2019 | .0741554 .0046992 15.78 0.000 .0649452 .0833657
age | -.0087547 .0020947 -4.18 0.000 -.0128602 -.0046492
|
industry |
2 | -.0992833 .0581482 -1.71 0.088 -.2132517 .0146851
3 | -.0152485 .0569085 -0.27 0.789 -.1267872 .0962902
-----
Note: dy/dx for factor levels is the discrete change from the base level.
.
.
end of do-file
.

```

PANEL DATA

```

. do "C:\Users\qclm20\AppData\Local\Temp\STD1404_000000.tmp"

. *PANEL
. xtset id year

Panel variable: id (strongly balanced)
Time variable: year, 2018 to 2020
Delta: 1 unit

. *POLS
. reg employment loan leverage age i.industry i.year

Source |      SS      df      MS      Number of obs   =      3,000
-----+-----
Model | 12420.9623      7 1774.42318   Prob > F      =      0.0000
Residual | 26350.2747    2,992    8.80691   R-squared      =      0.3204
-----+-----
Adj R-squared =      0.3188
Total | 38771.237    2,999   12.928055   Root MSE      =      2.9676

-----
employment | Coefficient Std. err.      t    P>|t|    [95% conf. interval]
-----+-----
loan | 5.400234 .2000421    27.00  0.000    5.008    5.792468
leverage | -.1839611 .2480157   -0.74  0.458   -.6702596 .3023375
age | .1002042 .0064168   15.62  0.000    .0876224 .112786
|
industry |
2 | 1.424489 .15558    9.16  0.000    1.119435 1.729544
3 | 2.390098 .1599179   14.95  0.000    2.076538 2.703658
|
year |
2019 | .6290588 .1328828    4.73  0.000    .3685078 .8896097
2020 | -1.440754 .1505381   -9.57  0.000   -1.735922 -1.145585
|
_cons | 5.34484 .2084468   25.64  0.000    4.936126 5.753553
-----

```

```
. *FE
. xtreg employment loan leverage age i.year, fe
note: 2020.year omitted because of collinearity.

Fixed-effects (within) regression              Number of obs   =       3,000
Group variable: id                            Number of groups  =       1,000

R-squared:                                    Obs per group:
Within = 0.1266                                min =           3
Between = 0.0003                               avg =          3.0
Overall = 0.0125                               max =           3

F(4,1996) = 72.30
corr(u_i, Xb) = -0.2330                      Prob > F = 0.0000
```

```
-----+-----
employment | Coefficient Std. err.      t    P>|t|    [95% conf. interval]
-----+-----
loan |    2.447373   .1678628    14.58  0.000    2.118168    2.776578
leverage |   -1.14714   .58879    -1.95  0.052   -2.301847    .0075678
age |   -.1014947   .0547248    -1.85  0.064   -.2088185    .0058291
|
year |
2019 |    .8373703   .0853104     9.82  0.000    .6700636    1.004677
2020 |           0 (omitted)
|
_cons |   10.4096    .9163146    11.36  0.000    8.612564    12.20663
-----+-----

sigma_u |   3.240942
sigma_e |   2.0655693
rho |   .71113776 (fraction of variance due to u_i)
-----+-----

F test that all u_i=0: F(999, 1996) = 4.89                      Prob > F = 0.0000
```

```
. *FD
. reg D.employment D.loan D.leverage i.year
```

```
Source |         SS            df          MS      Number of obs   =       2,000
-----+-----+-----+-----+-----+-----
Model |   1428.53224          3   476.177414      Prob > F          =       0.0000
Residual |  16821.8673        1,996   8.42778921      R-squared          =       0.0783
-----+-----+-----+-----+-----
Adj R-squared =       0.0769
Total |   18250.3995        1,999   9.12976463      Root MSE          =       2.9031
```

```
-----+-----
D.employment | Coefficient Std. err.      t    P>|t|    [95% conf. interval]
-----+-----
loan |
D1. |    2.185989   .1925808    11.35  0.000    1.808309    2.56367
|
leverage |
D1. |   -1.212884   .6727009    -1.80  0.072   -2.532153    .1063858
|
year |
```

2020		-1.583896	.146531	-10.81	0.000	-1.871266	-1.296526
_cons		.736327	.0919191	8.01	0.000	.5560596	.9165944

```

.
end of do-file
.

```

DiD : Parralel tends

```

. do "C:\Users\qclm20\AppData\Local\Temp\STD2b10_000000.tmp"
. preserve
. capture drop treated2020
. bysort id: egen treated2020 = max(year == 2020 & loan == 1)
.
. tempfile base tn un au
. save 'base', replace
(file C:\Users\qclm20\AppData\Local\Temp\ST_2b10_000002.tmp not found)
file C:\Users\qclm20\AppData\Local\Temp\ST_2b10_000002.tmp saved as .dta
format
.
. use 'base', clear
. keep if state == 1 & treated2020 == 1
(1,950 observations deleted)
. collapse (mean) treated_north = employment, by(year)

. save 'tn', replace
(file C:\Users\qclm20\AppData\Local\Temp\ST_2b10_000003.tmp not found)
file C:\Users\qclm20\AppData\Local\Temp\ST_2b10_000003.tmp saved as .dta
format
.
. use 'base', clear
. keep if state == 1 & treated2020 == 0
(2,550 observations deleted)
. collapse (mean) untreated_north = employment, by(year)
. save 'un', replace
(file C:\Users\qclm20\AppData\Local\Temp\ST_2b10_000004.tmp not found)
file C:\Users\qclm20\AppData\Local\Temp\ST_2b10_000004.tmp saved as .dta
format
.
. use 'base', clear

. keep if treated2020 == 0
(1,050 observations deleted)

. collapse (mean) all_untreated = employment, by(year)

. save 'au', replace
(file C:\Users\qclm20\AppData\Local\Temp\ST_2b10_000005.tmp not found)
file C:\Users\qclm20\AppData\Local\Temp\ST_2b10_000005.tmp saved as .dta
format

```

```

.
. use 'tn', clear

. merge 1:1 year using 'un', nogen

Result                                Number of obs
-----
Not matched                           0
Matched                               3
-----

. merge 1:1 year using 'au', nogen

Result                                Number of obs
-----
Not matched                           0
Matched                               3
-----

.
. list year treated_north untreated_north all_untreated, clean noobs

year   treaten_h   untreaen_h   all_unen_d
-----
2018   10.47429    6.566667    7.252308
2019    11.54      8.213333    7.798461
2020   12.87429    8.026667    6.958462

. * treated north vs untreated north
. twoway ///
> (connected treated_north year, sort msymbol(o)) ///
> (connected untreated_north year, sort msymbol(triangle)), ///
> title("treated north vs untreated north") ///
> xtitle("period") ///
> ytitle("mean employment") ///
> xlabel(2018 "end 2018" 2019 "end 2019" 2020 "end 2020") ///
> xline(2019, lpattern(dash) lcolor(red)) ///
> legend(order(1 "treated north" 2 "untreated north")) ///
> name(g_north, replace)

.
. * treated north vs all untreated
. twoway ///
> (connected treated_north year, sort msymbol(o)) ///
> (connected all_untreated year, sort msymbol(square)), ///
> title("treated north vs all untreated") ///
> xtitle("period") ///
> ytitle("mean employment") ///
> xlabel(2018 "end 2018" 2019 "end 2019" 2020 "fin 2020") ///
> xline(2019, lpattern(dash) lcolor(red)) ///
> legend(order(1 "treated north" 2 "all untreated")) ///
> name(g_alluntreated, replace)

.
. *combine

```

```

. graph combine g_north g_alluntreated, ///
> title("parallel trends inspection")
. restore
.
end of do-file
.

```

Match In All

```

. do "C:\Users\qcilm20\AppData\Local\Temp\STD2b10_000000.tmp"
. preserve
. sort id year
. *(activate this option to have north untreated only)
. *keep if state == 1
. gen employment_2019 = employment if year == 2019
(2,000 missing values generated)

. by id: egen emp19 = max(employment_2019)

. keep if year == 2020
(2,000 observations deleted)

. * outcome in 2020
. gen employment_2020 = employment

. * covariables in matching
. gen lev20 = leverage

. gen age20 = age

. gen ind20 = industry

. * traitement
. gen treated_north = (state == 1 & loan == 1)

. keep id state loan treated_north employment_2020 emp19 lev20 age20 ind20

. drop if missing(employment_2020, emp19, lev20, age20, ind20)
(0 observations deleted)

. * matching nearest neighbour
. teffects nnmatch (employment_2020 emp19 lev20 age20 i.ind20) (treated_north)
> , metric(mahalanobis) generate(matchid) nneighbor(1) atet ///
>

```

```

Treatment-effects estimation      Number of obs      =      1,000
Estimator      : nearest-neighbor matching      Matches: requested =      1
Outcome model  : matching                      min =      1
Distance metric: Mahalanobis                      max =      1
-----

```

```

|              AI robust
employm~2020 | Coefficient  std. err.      z    P>|z|      [95% conf. interval]

```



```

-----+-----
ATET      |
treated_no~h |
(1 vs 0) |      4.7      .286582      16.40      0.000      4.13831      5.26169
-----+-----

. *balance before matching
.
. ttest emp19, by(treated_north)

Two-sample t test with equal variances
-----+-----
Group |      Obs      Mean      Std. err.      Std. dev.      [95% conf. interval]
-----+-----
0 |      650      7.798462      .112811      2.876127      7.576943      8.01998
1 |      350      11.54      .1553241      2.905848      11.23451      11.84549
-----+-----
Combined |      1,000      9.108      .1072933      3.392911      8.897454      9.318546
-----+-----
diff |      -3.741538      .1913768      -4.117085      -3.365991
-----+-----
diff = mean(0) - mean(1)                                t = -19.5506
H0: diff = 0                                             Degrees of freedom =      998

Ha: diff < 0              Ha: diff != 0              Ha: diff > 0
Pr(T < t) = 0.0000        Pr(|T| > |t|) = 0.0000        Pr(T > t) = 1.0000

. ttest lev20, by(treated_north)

Two-sample t test with equal variances
-----+-----
Group |      Obs      Mean      Std. err.      Std. dev.      [95% conf. interval]
-----+-----
0 |      650      .3980914      .0092103      .2348185      .3800058      .4161771
1 |      350      .4273905      .0124848      .2335699      .4028356      .4519455
-----+-----
Combined |      1,000      .4083461      .0074213      .2346821      .393783      .4229092
-----+-----
diff |      -.0292991      .0155394      -.0597928      .0011946
-----+-----
diff = mean(0) - mean(1)                                t = -1.8855
H0: diff = 0                                             Degrees of freedom =      998

Ha: diff < 0              Ha: diff != 0              Ha: diff > 0
Pr(T < t) = 0.0298        Pr(|T| > |t|) = 0.0597        Pr(T > t) = 0.9702

. ttest age20, by(treated_north)

Two-sample t test with equal variances
-----+-----
Group |      Obs      Mean      Std. err.      Std. dev.      [95% conf. interval]
-----+-----
0 |      650      17.34462      .3287021      8.380291      16.69917      17.99006
1 |      350      17.58      .4587911      8.583195      16.67766      18.48234

```

```
-----+-----
Combined |   1,000   17.427   .2671592   8.448316   16.90274   17.95126
-----+-----
diff |           -.2353846   .5603489           -1.334982   .8642126
-----+-----
```

```
diff = mean(0) - mean(1)                                t =  -0.4201
H0: diff = 0                                             Degrees of freedom =    998
```

```
Ha: diff < 0          Ha: diff != 0          Ha: diff > 0
Pr(T < t) = 0.3373    Pr(|T| > |t|) = 0.6745    Pr(T > t) = 0.6627
```

```
. tab ind20 treated_north, chi2
```

```

|      treated_north
ind20 |          0          1 |      Total
-----+-----+-----
1 |      142      26 |      168
2 |      322     118 |      440
3 |      186     206 |      392
-----+-----+-----
Total |      650     350 |     1,000
```

```
Pearson chi2(2) = 94.1730  Pr = 0.000
```

```
. tebalance summarize emp19, baseline
```

```
Covariate balance summary
```

```
Raw      Matched
-----
Number of obs =      1,000      700
Treated obs   =        350      350
Control obs   =        650      350
-----
```

```
-----
|          Means          Variances
|      Control    Treated      Control    Treated
-----+-----
emp19 |   7.798462     11.54     8.272109   8.443954
-----
```

```
. tebalance density emp19
```

```
. tebalance density lev20
```

```
. tebalance density age20
```

```
. *****SCORE PROPENSITY
```

```
. logit loan lev20 emp19 age20 i.ind20, nolog
```

```
Logistic regression                                Number of obs = 1,000
LR chi2(5)    = 370.78
Prob > chi2   = 0.0000
```

Log likelihood = -462.05414 Pseudo R2 = 0.2863

loan	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
lev20	.7731715	.3545481	2.18	0.029	.0782701	1.468073
emp19	.4568571	.0338536	13.50	0.000	.3905053	.5232089
age20	-.0436706	.0102557	-4.26	0.000	-.0637715	-.0235698
ind20						
2	.1223457	.273503	0.45	0.655	-.4137103	.6584018
3	.9278118	.2719087	3.41	0.001	.3948806	1.460743
_cons	-5.067751	.4225714	-11.99	0.000	-5.895976	-4.239526

```
. teffects psmatch (employment_2020) (treated_north emp19 lev20 age20 i.ind20,
> logit), nneighbor(1) generate(ps) atet
```

```
Treatment-effects estimation      Number of obs      =      1,000
Estimator      : propensity-score matching      Matches: requested =      1
Outcome model  : matching                                min =      1
Treatment model: logit                                max =      2
```

AI robust						
employ~2020	Coefficient	std. err.	z	P> z	[95% conf. interval]	
-----+-----						
ATET						
treated_no~h						
(1 vs 0)	4.361429	.3778127	11.54	0.000	3.620929	5.101928

```
. *****
.
end of do-file
.
```

Match In North

```
. do "C:\Users\qclm20\AppData\Local\Temp\STD2b10_000000.tmp"
. preserve
. sort id year
. *(activate this option to have north untreated only)
. keep if state == 1
(1,500 observations deleted)

. gen employment_2019 = employment if year == 2019
(1,000 missing values generated)

. by id: egen emp19 = max(employment_2019)

. keep if year == 2020
```

```

(1,000 observations deleted)

. * outcome in 2020
. gen employment_2020 = employment

. * covariables in matching
. gen lev20 = leverage

. gen age20 = age

. gen ind20 = industry

. * traitement
. gen treated_north = (state == 1 & loan == 1)

. keep id state loan treated_north employment_2020 emp19 lev20 age20 ind20

. drop if missing(employment_2020, emp19, lev20, age20, ind20)
(0 observations deleted)

. * matching nearest neighbour
. teffects nnmatch (employment_2020 emp19 lev20 age20 i.ind20) (treated_north)
> , metric(mahalanobis) generate(matchid) nneighbor(1) atet ///
>

Treatment-effects estimation      Number of obs      =      500
Estimator      : nearest-neighbor matching      Matches: requested =      1
Outcome model  : matching                        min =      1
Distance metric: Mahalanobis                    max =      1
-----
|                AI robust
employm~2020 | Coefficient  std. err.      z    P>|z|      [95% conf. interval]
-----+-----
ATET          |
treated_no~h |
(1 vs 0) |    3.925714   .4117995    9.53   0.000    3.118602    4.732826
-----

. *balance before matching
.
. ttest emp19, by(treated_north)

Two-sample t test with equal variances
-----
Group |      Obs      Mean   Std. err.   Std. dev.   [95% conf. interval]
-----+-----
0 |      150   8.213333   .2060791   2.523944   7.806118   8.620548
1 |      350   11.54     .1553241   2.905848   11.23451   11.84549
-----+-----
Combined |      500   10.542     .1423833   3.183788   10.26226   10.82174
-----+-----
diff |           -3.326667   .2729647           -3.862971   -2.790362
-----
diff = mean(0) - mean(1)                                t = -12.1872

```

H0: diff = 0 Degrees of freedom = 498

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
 Pr(T < t) = 0.0000 Pr(|T| > |t|) = 0.0000 Pr(T > t) = 1.0000

. ttest lev20, by(treated_north)

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
0	150	.3629249	.0195083	.2389273	.3243762	.4014736
1	350	.4273905	.0124848	.2335699	.4028356	.4519455
Combined	500	.4080508	.0105902	.2368035	.387244	.4288577
diff		-.0644657	.0229518		-.1095599	-.0193714

diff = mean(0) - mean(1) t = -2.8087
 H0: diff = 0 Degrees of freedom = 498

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
 Pr(T < t) = 0.0026 Pr(|T| > |t|) = 0.0052 Pr(T > t) = 0.9974

. ttest age20, by(treated_north)

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
0	150	17.26	.695748	8.521138	15.88519	18.63481
1	350	17.58	.4587911	8.583195	16.67766	18.48234
Combined	500	17.484	.3826962	8.557348	16.73211	18.23589
diff		-.32	.8358267		-1.962181	1.322181

diff = mean(0) - mean(1) t = -0.3829
 H0: diff = 0 Degrees of freedom = 498

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
 Pr(T < t) = 0.3510 Pr(|T| > |t|) = 0.7020 Pr(T > t) = 0.6490

. tab ind20 treated_north, chi2

		treated_north		
ind20	0	1	Total	
1	16	26	42	
2	76	118	194	
3	58	206	264	
Total	150	350	500	

Pearson chi2(2) = 17.1946 Pr = 0.000

. tebalance summarize emp19, baseline

Covariate balance summary

Raw Matched

```
-----
Number of obs =      500      700
Treated obs   =      350      350
Control obs   =      150      350
-----
```

```
-----
|           Means           Variances
|   Control   Treated   Control   Treated
-----+-----
emp19 |   8.213333    11.54    6.370291   8.443954
-----
```

. tebalance density emp19

. tebalance density lev20

. tebalance density age20

. *****SCORE PROPENSITY

. logit loan lev20 emp19 age20 i.ind20, nolog

```
Logistic regression                Number of obs =      500
LR chi2(5)      = 164.77
Prob > chi2     = 0.0000
Log likelihood = -223.04791          Pseudo R2      = 0.2697
```

```
-----
loan | Coefficient   Std. err.      z    P>|z|    [95% conf. interval]
-----+-----
lev20 |   1.617839    .5119727     3.16   0.002    .6143914    2.621288
emp19 |   .5132631    .0537713     9.55   0.000    .4078733    .6186528
age20 |  -.0614062    .0155395    -3.95   0.000   -.091863   -.0309495
|
ind20 |
2 |  -.7163304    .4225629    -1.70   0.090   -1.544539    .1118777
3 |  -.1215784    .42092     -0.29   0.773   -.9465665    .7034097
|
_cons |  -3.425377    .6089048    -5.63   0.000   -4.618809   -2.231946
-----
```

. teffects psmatch (employment_2020) (treated_north emp19 lev20 age20 i.ind20,
> logit), nneighbor(1) generate(ps) atet

```
Treatment-effects estimation      Number of obs      =      500
Estimator      : propensity-score matching      Matches: requested =      1
Outcome model  : matching                                min =      1
```

```

Treatment model: logit                                max =      1
-----
|               AI robust
employm~2020 | Coefficient  std. err.      z    P>|z|    [95% conf. interval]
-----+-----
ATET          |
treated_no~h |
(1 vs 0) |    3.377143   .5112141    6.61  0.000    2.375182    4.379104
-----

. *****
.
end of do-file

```

DiD, DiD+Match

```

. *DiD
. regress d.employment d.loan if year==2020

Source |         SS            df          MS      Number of obs   =       1,000
-----+-----
Model |   1075.51043           1   1075.51043    Prob > F           =       0.0000
Residual |   8641.24857          998    8.6585657    R-squared           =       0.1107
-----+-----
Total |    9716.759           999    9.72648549   Root MSE           =       2.9425
-----

D.employment | Coefficient  Std. err.      t    P>|t|    [95% conf. interval]
-----+-----
loan |
D1. |    2.174286   .1950888    11.15  0.000    1.791454    2.557117
|
_cons |         -.84   .1154161    -7.28  0.000   -1.066486   -.613514
-----

. regress d.employment d.loan d.leverage if year==2020

Source |         SS            df          MS      Number of obs   =       1,000
-----+-----
Model |   1093.53604           2    546.768019   Prob > F           =       0.0000
Residual |   8623.22296          997    8.64917047   R-squared           =       0.1125
-----+-----
Total |    9716.759           999    9.72648549   Root MSE           =       2.9409
-----

D.employment | Coefficient  Std. err.      t    P>|t|    [95% conf. interval]
-----+-----
loan |
D1. |    2.187839   .1952088    11.21  0.000    1.804771    2.570906
|
leverage |
D1. |   -1.40452   .9729044    -1.44  0.149   -3.313695    .5046554

```

```

|
_cons | -.8487649 .1155131 -7.35 0.000 -1.075442 -.6220882
-----+-----

. * matching + did
. *clear
. xtset id year

Panel variable: id (strongly balanced)
Time variable: year, 2018 to 2020
Delta: 1 unit

. regress d.employment d.loan if year==2020

Source |      SS      df      MS      Number of obs =      1,000
-----+-----+-----+-----+-----+-----
                F(1, 998)      =      124.21
Model |  1075.51043      1  1075.51043  Prob > F      =      0.0000
Residual |  8641.24857     998   8.6585657  R-squared      =      0.1107
-----+-----+-----+-----+-----
                Adj R-squared =      0.1098
Total |   9716.759     999   9.72648549  Root MSE      =      2.9425
-----+-----

D.employment | Coefficient Std. err.      t      P>|t|      [95% conf. interval]
-----+-----+-----+-----+-----+-----
loan |
D1. |   2.174286   .1950888    11.15   0.000    1.791454    2.557117
|
_cons |    -.84   .1154161    -7.28   0.000   -1.066486   -.613514
-----+-----

. regress d.employment d.loan d.leverage if year==2020

Source |      SS      df      MS      Number of obs =      1,000
-----+-----+-----+-----+-----
                F(2, 997)      =      63.22
Model |  1093.53604      2   546.768019  Prob > F      =      0.0000
Residual |  8623.22296     997   8.64917047  R-squared      =      0.1125
-----+-----+-----+-----+-----
                Adj R-squared =      0.1108
Total |   9716.759     999   9.72648549  Root MSE      =      2.9409
-----+-----

D.employment | Coefficient Std. err.      t      P>|t|      [95% conf. interval]
-----+-----+-----+-----+-----+-----
loan |
D1. |   2.187839   .1952088    11.21   0.000    1.804771    2.570906
|
leverage |
D1. |  -1.40452   .9729044    -1.44   0.149   -3.313695    .5046554
|
_cons |  -.8487649   .1155131    -7.35   0.000   -1.075442   -.6220882
-----+-----

.
. * matching + did in 2020
. preserve

```



```

. sort id year
. capture drop employment_2019 emp19 change_employment lev20 age20 ind20 ///
> treated_north obs2020 matchid* matched_treated matched_control matched_sample
.
. gen employment_2019 = employment if year == 2019
(2,000 missing values generated)
. by id: egen emp19 = max(employment_2019)
.
. keep if year == 2020
(2,000 observations deleted)
. * keep if state == 1
. gen change_employment = employment - emp19
. gen lev20 = leverage
. gen age20 = age
. gen ind20 = industry
. gen treated_north = (state == 1 & loan == 1)
.
. keep id treated_north change_employment emp19 lev20 age20 ind20
. drop if missing(change_employment, emp19, lev20, age20, ind20)
(0 observations deleted)
. gen obs2020 = _n
.
. teffects nnmatch ///
> (change_employment emp19 lev20 age20 i.ind20) ///
> (treated_north), ///
> metric(mahalanobis) nneighbor(1) atet generate(matchid)

Treatment-effects estimation      Number of obs      =      1,000
Estimator      : nearest-neighbor matching      Matches: requested =      1
Outcome model  : matching                        min =      1
Distance metric: Mahalanobis                    max =      1
-----
|               AI robust
change_empl`t | Coefficient  std. err.      z    P>|z|      [95% conf. interval]
-----+-----
ATET          |
treated_north |
(1 vs 0) |      4.24   .3083399   13.75   0.000    3.635665   4.844335
-----

.
. tempfile matchbase treatedids controlids counts

. save 'matchbase', replace
(file C:\Users\qclm20\AppData\Local\Temp\ST_c0c_000002.tmp not found)
file C:\Users\qclm20\AppData\Local\Temp\ST_c0c_000002.tmp saved as .dta format
. use 'matchbase', clear
. keep if treated_north == 1
(650 observations deleted)
. keep id
. duplicates drop
Duplicates in terms of all variables
(0 observations are duplicates)
. gen matched_treated = 1

```

```

. save 'treatedids', replace
(file C:\Users\qclm20\AppData\Local\Temp\ST_c0c_000003.tmp not found)
file C:\Users\qclm20\AppData\Local\Temp\ST_c0c_000003.tmp saved as .dta format
. use 'matchbase', clear
. keep if treated_north == 1
(650 observations deleted)
. keep matchid1
. rename matchid1 obs2020
. drop if missing(obs2020)
(0 observations deleted)
. duplicates drop
Duplicates in terms of all variables
(174 observations deleted)
.
. merge m:1 obs2020 using 'matchbase', nogen keep(match) keepusing(id)
(variable obs2020 was long, now double to accommodate using data's values)

```

Result	Number of obs
Not matched	0
Matched	176

```

. keep id
. duplicates drop
Duplicates in terms of all variables
(0 observations are duplicates)
. gen matched_control = 1
. save 'controlids', replace
(file C:\Users\qclm20\AppData\Local\Temp\ST_c0c_000004.tmp not found)
file C:\Users\qclm20\AppData\Local\Temp\ST_c0c_000004.tmp saved as .dta format
.
. restore

. * full panel
. merge m:1 id using 'treatedids', nogen keep(master match)

```

Result	Number of obs
Not matched	1,950
from master	1,950
from using	0
Matched	1,050

```

. replace matched_treated = 0 if missing(matched_treated)
(1,950 real changes made)

. merge m:1 id using 'controlids', nogen keep(master match)

```

Result	Number of obs
Not matched	2,472

```

from master                2,472
from using                  0

Matched                    528
-----

. replace matched_control = 0 if missing(matched_control)
(2,472 real changes made)

. gen matched_sample = .
(3,000 missing values generated)

. replace matched_sample = 1 if matched_treated == 1
(1,050 real changes made)
. replace matched_sample = 0 if matched_control == 1
(528 real changes made)

. label define ms 1 "matched treated" 0 "matched controls", replace

. label values matched_sample ms

. * keep some variables
. keep id year employment matched_sample state

. keep if matched_sample < .
(1,422 observations deleted)
.
. preserve
. keep id matched_sample

. duplicates drop

Duplicates in terms of all variables

(1,052 observations deleted)
. contract matched_sample
. label values matched_sample ms
. list matched_sample _freq, noobs
+-----+
| matched_sample _freq |
+-----+
| matched controls    176 |
| matched treated    350 |
+-----+

. restore
. * graph of tends
. preserve
. collapse (mean) mean_emp = employment, by(year matched_sample)
. twoway ///
> (connected mean_emp year if matched_sample == 1, sort msymbol(o)) ///
> (connected mean_emp year if matched_sample == 0, sort msymbol(square)), ///
> title("matched treated vs matched controls") ///
> xtitle("period") ///
> ytitle("mean employment") ///

```

```

> xlabel(2018 "end 2018" 2019 "start 2020" 2020 "end 2020", labsize(small)) ///
> xline(2019, lpattern(dash) lcolor(red)) ///
> xscale(range(2018 2020)) ///
> legend(order(1 "matched treated" 2 "matched controls")) ///
> name(g_match_did, replace)
. graph display g_match_did
. restore
end of do-file
.

```